MIL-W-37060A
31 March 1976
SUPERSEDING
MIL-W-37060
29 May 1974

MILITARY SPECIFICATION

WATER BATH, ELECTRIC, Serological, 115 volt, 50/60 Hz, AC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of constant temperature serological water bath suitable for use in laboratory procedures, having a temperature range from slightly above ambient to 65° Centigrade (°C.) when the bath is uncovered, and to 100°C. when the bath is covered.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Boxes, Shipping, Fiberboard.

Military

MIL-M-7298	Manual, Technical, Commercial Equipment.
MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

FSC 6640

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI C73-11	Dimensions of Plugs and Receptacles-General Purpose, 125 volts, 15 Amperes, 2-Pole, 3-Wire, Grounding Type.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

UNDERWRITERS' LABORATORIES, INC.

UL 499	Heating Appliances, Electric.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., Walt Whitman Road, Melville, L.I., NY 11746; 207 East Ohio Street, Chicago, IL 60611; and 1655 Scott Boulevard, Santa Clara, CA 95050.)

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UNITED STATES DEPARTMENT OF COMMERCE

National Bureau of Standards (NBS)

(Application for copies shall be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS

3.1 Fire and casualty hazards.

3.1.1 The contractor shall submit proof, prior to delivery, that the items to be delivered under the contract conform to the applicable requirements of UL 499. Proof shall consist of one of the following:

a. UL label or UL listing of the product, provided the contractor agrees to have Underwriters' Laboratories, Inc. forward a copy of the UL report directly to the contracting officer, upon the latter's request.

b. Laboratory report from a qualified independent laboratory. If an independent laboratory is used, it must be adequately equipped and competent to determine compliance with UL standards. The qualification of such laboratory shall be determined by the Government which reserves the right to require a facilities report and to conduct a facilities survey. Any laboratory report submitted shall include actual test methods, data taken and results of tests. Detailed findings of examinations necessary to determine compliance shall also be included. The test report shall also have a section that describes the test sample in such a way that the report can be used to insure identity of production items with the test sample. This section shall be supplemented with construction information such as materials used, spacings between electrical components, types of bushings, photographs, wiring diagrams, sketches and parts list, all appropriately identified.

3.1.2 Compliance with above paragraph does not excuse compliance with all other requirements of the contract and specifications.

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3.2 Materials. Materials used shall be suitable for intended uses, and as hereinafter specified.

3.3 Design. The water bath shall consist essentially of the following:

a. Housing with an insulated water chamber
b. Heater
c. Thermostatic temperature control
d. Pilot light
e. On-off line switch
f. Power cord
g. Cover
h. Thermometer

* 3.3.1 Housing. The housing shall be of double-wall construction. The water chamber (tank) shall be formed by the inner walls and inside bottom of the housing. The water chamber shall be of welded construction and shall be fabricated of suitable corrosion-resisting steel. The water chamber shall be leakproof, and shall be free of holes or openings that can result in leakage of the water. The water chamber shall have minimum dimensions (inches) of 11-15/16 wide by 6-7/16 deep by 4-5/8 high. A thermometer holder shall be securely attached to the housing in order to accommodate the thermometer specified in 3.3.8. The housing shall have sufficient thickness to be free of buckling and bulging. The housing shall provide space for the insulation, internal wiring, heater(s) and electric controls. Exposure of the insulation and open seams on the housing shall not be allowed. All seams shall be properly welded, soldered, or brazed. The bottom corners of the housing shall be supplied with projecting feet. The maximum dimensions of the outside of the housing without projections (e.g. feet, electric controls) shall be 16-1/8 wide by 10-9/16 deep by 10-5/16 high (inches). The top of the housing shall be designed to receive the cover specified in 3.3.7. The use of a gasket on the top of the housing to accommodate the cover shall be acceptable.

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3.3.2 Heater. The heater element(s) shall be suitable for operation on 115 volt, 50/60 Hz, AC supply lines, and shall not require more than 375 watts (total) for proper operation. The heater operation shall be controlled by the thermostatic temperature control (see 3.3.3) and the on-off line switch (see 3.3.5).

* 3.3.3 Thermostatic temperature control. The thermostatic temperature control shall be capable of maintaining the water bath at any temperature within the range to an accuracy (uniformity) of $\pm$ 0.5°C. (see 3.6.2). The thermostatic temperature control shall have a dial control for selection of the required temperature of the water bath. The dial control shall be attached to the front of the housing and shall include reference markings.

3.3.4 Pilot light. A pilot light shall be attached to the front of the housing, and shall indicate when the heater element(s) (see 3.3.2) are energized.

3.3.5 On-off line switch. An on-off line switch shall be attached to the front of the housing, and shall control all electric power to water bath apparatus. The "on" and "off" positions of the switch shall be marked on the switch or housing.

* 3.3.6 Power cord. A 3-conductor power cord with non-detachable connector shall be supplied. A strain relief bushing shall be provided where the cord enters the housing. The connector plug shall be a 3-blade male plug described in ANSI C73-11. The overall length of cord external to the housing shall be not less than 6 feet.

* 3.3.7 Cover. A corrosion-resisting metal cover or a transparent plastic cover shall be supplied. The top of the cover shall have a plastic handle. The cover shall be capable of covering the top of the water chamber without obstructing the thermometer (see 3.3.8).

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* 3.3.8 Thermometer. The thermometer shall be a partial immersion mercury-in-glass type, and shall have approximate dimensions of 11 inches long and 1/4 inch diameter. The thermometer shall include a 35-100°C. range with 1° graduations. The thermometer shall have markings to clearly indicate the proper depth of immersion for the thermometer in the water bath. The thermometer shall be capable of being inserted and positioned in the thermometer holder attached to the housing. When tested as specified in 4.4.1, the thermometer shall have an accuracy of ± 0.75°C. at center scale and ± 1.5°C. at the ends of the scale.

3.4 Finish. The exposed surfaces of the water bath which are not corrosion-resisting steel shall have a protective finish against corrosion. The finish shall be free of scratches, chipping or blisters.

3.5 Workmanship. Workmanship shall be first class in every respect. The water bath shall be free from defects which detract from its appearance or may impair its serviceability.

3.6 Performance.

3.6.1 Dielectric strength. There shall be no electrical breakdown when the water bath is tested as specified in 4.4.2.

* 3.6.2 Temperature control. When tested as specified in 4.4.3, the temperature controller shall maintain the uniformity of the bath temperatures within ± 0.5°C.

3.7 Service data. The contractor shall furnish with each unit two copies each of a technical manual which contains complete instructions for installation, operation and maintenance, ahd lists of component parts. The manual shall comply with the requirements of MIL-M-7298. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the contracting officer, via the cognizant Government inspector, two preliminary copies of his proposed technical manual, for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary manuals shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of

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30 days for such review) and to allow incorporation by the contractor of any required revisions and corrections in the final manual without delaying delivery under the contract or purchase order. In addition, the contractor shall furnish 15 copies of the approved technical manual to the contracting officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of contract and propose to furnish manuals identical with those previously accepted, the requirements for review of drafts and submission of 15 additional copies may be waived upon request to the contracting officer. Waiver of submission of 15 additional copies shall be at the option of the Government.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with Table I and MIL-STD-105.

* Table I. For examination

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	2.5
Dimensional examination	S-3	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with Table II and MIL-STD-105.

Table II. For testing

Component (unit of product)	Characteristic	Requirement	Test procedure	Inspection level
Thermometer	Accuracy	3.3.8	4.4.1	S-1*
Water bath	Dielectric strength	3.6.1	4.4.2	100%
Water bath	Temperature control	3.6.2	4.4.3	S-2*

* Acceptance number shall be zero.

4.3 Examination. The water bath shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the classification of defects shown in Table III. Examination shall not be restricted to the classified possible defects listed below:

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Table III. Classification of defects

Categories	Defects
Major	
101	Unit not complete with all component parts.
102	Unit not free of cracks, tears, dents or distortion.
103	Unit not free of wobbling or rocking when placed on a flat surface.
104	Unit not free of sharp edges or sharp burrs.
105	Line cord not adequately secured within housing.
106	Line cord not free of cuts, tears or abrasion.
107	Blades of line cord plug not properly aligned.
108	Components not usable or not capable of being properly secured to unit.
109	Unit not free of corrosion.
Minor	
201	Electric control markings not complete, legible or permanent.
202	Finish not free of scratches, chipping or blisters.
203	Unit not free of dirt, grease or foreign matter.

4.3.2 Dimensional examination. The water bath shall be examined for defects in dimensions. Any dimension that is not within specified limits shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Thermometer accuracy. The thermometer shall be tested against a standard thermometer at 35°C., 67°C. and 100°C. Calibration of the standard thermometer shall be traceable to the National Bureau of Standards.

4.4.2 Dielectric strength. With the line plug disconnected and actuating switch(es) deactivated, the water bath shall be subjected to a 60 Hz AC potential, gradually increasing to 900 volts, applied between the current-carrying parts and the metal enclosure and maintained at maximum value for a period of 1 minute.

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4.4.3 Temperature control. The water bath and thermometer shall be set up in accordance with the service data instructions of 3.7. With the tank filled with water and the unit connected to a power source, the temperature control shall be adjusted for the following bath operating temperatures:

Uncovered bath---38°C. ± 2°, 56°C. ± 2°, and 65°C. minimum.

Covered bath-----38°C. ± 2°, 56°C. ± 2°, and boiling water.

After each specified bath temperature has been reached, 10 thermometer temperature readings shall be recorded at 1-minute intervals.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One water bath, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit preparation. The thermometer shall be packaged in a plastic or spiral-wound chipboard container of suitable size. The container shall include suitable interior fittings or shall contain sufficient cushioning material to prevent damage to the thermometer. The thermometer along with technical manuals shall be placed and secured within the water bath chamber. The plastic cover may be packaged in a fiberboard box of appropriate size or shall be secured in place on the water bath. The line cord shall be neatly coiled and the coil secured. The water bath shall be wrapped in kraft paper or other suitable material and cushioned with corrugated fiberboard or other suitable material to insure against marring, scratching or damage to the unit.

5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

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5.2 Packing.

* 5.2.1 Level A. Each unit, prepared as specified in Level A above, shall be packed in an exterior container designed for a type 3 load and constructed in accordance with PPP-B-601, overseas type; PPP-B-621, class 2, or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed in accordance with the appendix of PPP-B-636.

* 5.2.2 Level B. Each unit prepared as specified in Packaging Level A shall be packed in an exterior container designed for a type 3 load and constructed in accordance with PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking. Exterior container shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. National Stock Number (NSN).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.2 The item covered by this specification is listed in the Federal Supply Catalog as follows:

National Stock No.	Item Identification
6640-00-145-1158	WATER BATH, ELECTRIC, Serological, 115 volt, 50/60 Hz, AC

6.3 This specification does not include all types, classes, grades or sizes of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, classes, grades and sizes normally procured to meet Military requirements.

6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6640-0922

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DOCUMENT IDENTIFIER AND TITLE WATER
MIL-W-37060A BATH, ELECTRIC, Serological

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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☐ YES ☐ NO *(If "Yes", in what way?)*

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